

Abena Boadi-Agyemang

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 Pittsburgh, PA, USA

Research Interests

I am a mixed-methods researcher who uses human-centered & participatory design approaches to designing AI agents & robots that can support people with disabilities (PwDs) in their daily lives. I focus on social agents to promote human mental well-being in people with chronic anxiety & physically assistive & service robots to support PwDs & their communities.

Keywords – human-robot interaction, human-AI interaction, human-centered design, accessibility, social robots, assistive agents, service robots

Skills

robotics | programming: python, javascript, web APIs (REST, Fetch)
embedded hardware: mechanical design (Solidworks, Fusion 360), circuit design, embedded C
reinforcement learning: model-based & model-free; multi-arm bandits

user research | usability: A/B testing, think aloud, contextual inquiry;
concept testing (Figma, Adobe Xd); **analysis:** thematic, affinity diagramming; **graphics & visual** (Unity 3d, processing)

community design | participatory workshops: facilitating, brainstorming, prototyping;
multi-stakeholder: value-sensitive design; **inclusive design:** mixed dis/ability workshops

human research | experimental studies, statistical analysis (JMP, Matlab, R); interview, focus group, ethnography, survey

Education

Carnegie Mellon University | Ph.D. in Robotics, School of Computer Science
August 2021 - present, Expected June 2026, Pittsburgh, PA, USA

- **Focus:** Human-Robot Interaction, Accessibility, Design Research
- **Committee:** Dr. Aaron Steinfeld (Chair), Dr. Jean Oh, Dr. Patrick Carrington, Dr. Cynthia L. Bennett (Google)

Carnegie Mellon University | M.S. in Robotics, School of Computer Science
August 2021 - May 2025, Pittsburgh, PA, USA

- **Focus:** Human-Robot Interaction, Accessibility, Design Research
- **Committee:** Dr. Aaron Steinfeld (Chair), Dr. Henny Admoni, Dr. Nikolas Martelaro, Michelle Zhao (Ph.D. Candidate)
- **Thesis:** “*Simulated Encounters of the Third Kind: Scenario-Based Approach to Designing Guide Robots*”

Stanford University | B.S. in Mechanical Engineering, School of Engineering
September 2017 - June 2021, Stanford, CA, USA

- **Focus:** Dynamic Systems & Controls
- **Capstone:** “*Using a Vibrotactile Stimulation Device to Improve Knee Extension During Stance Phase in Children with Spastic Cerebral Palsy*” (Teammates: Perez, G. A., Mhatre, K., Wang, J.) **[Nominated for Hoefer Prize for Writing in the Major]**

Experience

Robotics Institute, Carnegie Mellon University | Graduate Research Assistant
Aug 2021 - present, Pittsburgh, PA, USA

- Cooperative design workshops with non-roboticist people with disabilities (PwDs) for assistive & service robots focused on: brainstorming & prototyping robots for 3 types of assistive/service roles: (1) navigation support; (2) on-demand delivery; (3) emotional wellbeing.
- Designing experimental studies to improve robot adoption for human-robot assistive & collaborative tasks.

SHAPE Lab, Stanford University | Undergraduate Research Assistant
Jun 2019 - Jun 2021, Stanford, CA, USA

- Supervised by Dr. Sean Follmer. Mentored by Drs. Alexa F. Siu & Elyse D.Z. Chase.

- Designed & fabricated haptic devices to make GUIs more accessible for blind & low vision (BLV) users & provide non-visual adapted materials for BLV students.

Autodesk | Fusion Catalyst Student Intern

Jul 2020 - Dec 2020, Stanford, CA, USA

- Secured & oversaw community partnerships between Autodesk & three Stanford student organizations. Provided free Fusion 360 (CAD & CAM software) training to students.

VMware Women's Leadership Innovation Lab | Undergraduate Research Assistant

Oct 2018 - Jun 2019, Stanford, CA, USA

- Supervised by Dr. Melissa Abad. Conducted qualitative analysis of the gendered-language used in fortune 500 company leadership statements to further examine their effects on the work environment.
- Produced database of commonly-used words (by frequency) to inform inclusivity interventions for fortune 500 partners.

Graduate School of Education | Research Assistant

Nov 2017 - Mar 2018, Stanford, CA, USA

- Supervised by Dr. John Willinsky. Inspected & catalogued 20% of journals (on Beall's list) that use open source publishing software, Open Journals System & are classified as "predatory" (i.e., exploitative academic publishing).
- Prepared contact database of journal board members to solicit participation in interventions that improve their journal integrity.

Academic Publications

(* denotes equal contribution. [Google Scholar](#) for full papers)

1. **[Conference] Boadi-Agyemang, A.**, Moharana, S., Bennett, C.L., Carter, E.J., Carrington, P., Steinfeld, A. Pluriversal Approach to Co-Designing Delivery Robots with People with Disabilities. *Under Review*.
2. **[Journal] Carter, Lopez, Sun, Warriar, Wu, Ng, Wang, Boadi-Agyemang, Tomasic, Zimmerman, and Steinfeld.** Designing Future Smartphone Apps for Navigation, Exploration, and Object Identification with Blind Users. *Under Review*.
3. **[Journal] Shih, K.*, Boadi-Agyemang, A.***, Carter, E.J., Steinfeld, A. (2025). A Multi-Method Investigation of Guide Robot Characteristics for Blind and Low Vision Users. In *ACM THRI*.
4. **[Conference] Boadi-Agyemang, A.**, Schaldenbrand, P., Misra, V., Carter, E.J., Oh, J., Steinfeld, A. (2025). If I Move, Do You Move? Investigating the Role of Interpersonal Synchrony in Human-Robot Joint Painting. In *IEEE RO-MAN*.
5. **[Workshop Paper] Boadi-Agyemang, A.***, Moharana, S.* (2025). The Case for Em(body)ed Approaches to Disability-Centered Interaction Design. "Body Politics: Unpacking Tensions and Future Perspectives for Body-Centric Design Research in HCI" at *ACM CHI 2025*.
6. **[Short Paper] Boadi-Agyemang, A.*** and Park, M.* (2024). Designing Interactive Agents to Support Emotion Regulation in the Workplace through Guided Art-Making. In *Companion at 2024 ACM/IEEE HRI*.
7. **[Conference] Han, V. Y., Boadi-Agyemang, A.**, Lin, Y., Lindlbauer, D., and Ion, A. (2023). Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices. In *ACM UIST*.
8. **[Workshop Paper] Boadi-Agyemang, A.***, Shih, K.* and Steinfeld, A. (2023). Discovering User Needs and Preferences for Guide Robots: Challenges and Preliminary Insights. "Assistive Robots for Citizens" at *IEEE/RSJ IROS*.
9. **[Short Paper] Boadi-Agyemang, A.**, Carter, E.J., Siu, A. F., Steinfeld, A., and Martinez, M.O. (2023). Understanding Experiences, Attitudes, and Perspectives towards Designing Interactive Creative Tools for Teachers of Visually Impaired Students. In *ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.
10. **[Book] Siu, A. F., Chase, E. D. Z., Kim, G. S. H., Boadi-Agyemang, A.**, Gonzalez, E. J., and Follmer, S. (2021). Haptic Guidance to Support Design Education and Collaboration for Blind and Visually Impaired People. *Springer*.
11. **[Short Paper] Kim, L. H.*, Boadi-Agyemang, A.***, Siu, A. F., and Tang, J. (2020). When to Add Human Narration to Photo- Sharing Social Media. In *ASSETS*.
12. **[Short Paper] Chase, E. D. Z., Siu, A. F., Boadi-Agyemang, A.**, Kim, G. S. H., Gonzalez, E. J., and Follmer, S. (2020). PantoGuide: A Haptic and Audio Guidance System to Support Tactile Graphics Exploration. In *ASSETS*.

Select Awards

- **Carnegie Mellon University Uber Presidential PhD Fellowship ('22-'23)**
- **Cadence Diversity in Technology Scholarship ('23):** Students who exemplifies leadership & demonstrates a passion for technology.
- **University of Washington (UW) College of Engineering Dean's Fellowship [*declined*] ('21):** For top incoming doctoral applicants **(\$10k/yr)**.
- **UW Top Scholar Fellowship [*declined*], 2021:** to attract outstanding students **(\$2.5k)**.
- **Access_&Design Fellowship ('21):** for emerging BIPOC designers.
- **Botswana Top Achievers Award and Scholarship ('17-'21):** awarded by Botswana's Ministry of Education.
- **Stanford Mechanical Engineering Summer Undergraduate Research Institute (SURI) ('19, '20):** fellowship to conduct a ten-week original research project **(\$7.5k)**.
- **Academic Award of Excellence('18, '19):** awarded by Stanford University's Black Community Services Center.